

→ K Passes are useful

3.) SELECTION SORT -

Ex: A

8	6	3	2	5	4
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Ist Pass $k \rightarrow 8 \leftarrow i$
 6
 3
 2
 5
 4
 Find min ele.

IInd Pass $\textcircled{2}$ $\leftarrow i$ Sorted
 6
 3
 8
 5
 4
 Same Movement of k & j

IIIrd Pass

2	2
3	3
$\textcircled{6}$	4
8	8
5	5
4	6

 IVth Pass

2
3
4
$\textcircled{5}$
8
6

Vth Pass

2
3
4
5
6
8

 $\rightarrow$ We get sorted after 5 passes

Passes Required = $n-1$

No. of Comparisons = $\frac{n(n-1)}{2}$

No. of Swaps = $n-1 = O(n)$

Only algorithm that sorts with minimum no. of swaps.

~~It is not adaptive~~ It is not adaptive

Time complexity is always $O(n^2)$

Space complexity is $O(1)$.

It is not stable.